Adding & Subtracting Algebraic Fractions

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Written by: Naomi C
Reviewed by: Dan Finlay
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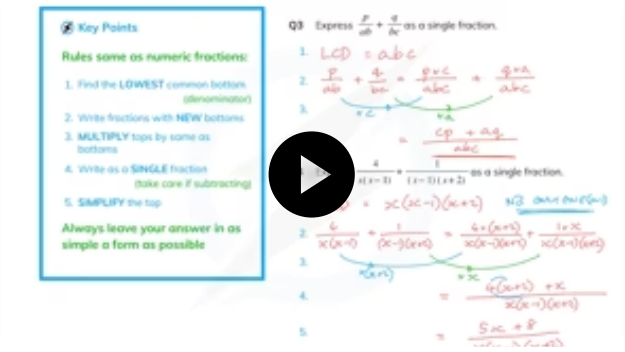

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Adding & subtracting algebraic fractions

How do I add (or subtract) two algebraic fractions?

- The **rules** for adding and subtracting **algebraic fractions** are the **same** as they are for **fractions with numbers**
- **STEP 1**
  Find the **lowest common denominator** (LCD)
  - Sometimes the LCD can be found by **multiplying** the denominators together
    - E.g. The LCD for the fractions $\frac{1}{x+2}$ and $\frac{1}{x+5}$ is $(x+2)(x+5)$
    - Similarly, with numbers, the LCD of $\frac{1}{2}$ and $\frac{1}{5}$ is $2 \times 5 = 10$
  - Although multiplying the denominators will always give you a multiple, it is **not necessarily the lowest** multiple
    - E.g. The LCD for the fractions $\frac{1}{x}$ and $\frac{1}{2x}$ is $2x$ (not $2x^2$) as both terms already include an x

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- Other examples include:
  - The LCD of $\frac{1}{x+2}$ and $\frac{1}{(x+2)(x-1)}$ is $(x+2)(x-1)$
  - The LCD of $\frac{1}{x+1}$ and $\frac{1}{(x+1)^2}$ is $(x+1)^2$
  - The LCD of $\frac{1}{(x+3)(x-1)}$ and $\frac{1}{(x+4)(x-1)}$ is $(x+3)(x-1)(x+4)$

- **STEP 2**
  Write each fraction over the **lowest common denominator**
  Multiply the numerator of each fraction by the **same amount** as the denominator
  - E.g. $\frac{x}{x-4}+\frac{1}{x+2}=\frac{x(x+2)}{(x-4)(x+2)}+\frac{(x-4)}{(x-4)(x+2)}$

- **STEP 3**
  Write as a **single fraction** over the lowest common denominator and **simplify the numerator**
  - Do this by **adding** or **subtracting** the **numerators**
  - Take particular care if subtracting
  - E.g.
    $\frac{x(x+2)+(x-4)}{(x-4)(x+2)}=\frac{x^2+2x+x-4}{(x-4)(x+2)}=\frac{x^2+3x-4}{(x-4)(x+2)}$

- **STEP 4**
  Check at the end to see if the top **factorises** and the fraction can be **simplified**
  - E.g. $\frac{(x+4)(x-1)}{(x-4)(x+2)}$, the top factorises but there are no common factors so it is in its most simple form

Examiner Tips and Tricks

Leaving the top and bottom of your answer in **factorised form** will help you see if anything cancels at the end.

Worked Example

(a) Express $\frac{x}{x+4}-\frac{3}{x-1}$ as a single fraction.

Answer:

The lowest common denominator is $(x+4)(x-1)$
Write each fraction over this common denominator, remember to multiply the top of the fractions too

$$\frac{x(x-1)}{(x+4)(x-1)}-\frac{3(x+4)}{(x-1)(x+4)}$$

Combine the fractions, as they now have the same denominator

Simplify the numerator
Be careful expanding with the negative signs

$$\frac{(x^2-x)-(3x+12)}{(x+4)(x-1)}=\frac{x^2-4x-12}{(x+4)(x-1)}$$

Factorise the top

$$\frac{(x+2)(x-6)}{(x+4)(x-1)}$$

There are no terms which would cancel here, so this is the final answer

$$\frac{(x+2)(x-6)}{(x+4)(x-1)}$$

(b) Express $\frac{x-4}{2(x-3)}-\frac{x-1}{2x}$ as a single fraction.

Answer:

The lowest common denominator is $2x(x-3)$
(You could also use $4x(x-3)$ but this wouldn't be the *lowest* common denominator)

Write each fraction over this common denominator, remember to multiply the top of the fractions too

$$\frac{x(x-4)}{2x(x-3)}-\frac{(x-1)(x-3)}{2x(x-3)}$$

Combine the fractions, as they now have the same denominator

$$\frac{x(x-4)-(x-1)(x-3)}{2x(x-3)}$$

Simplify the numerator
Be careful expanding with negative signs

$$\frac{(x^2-4x)-(x^2-4x+3)}{2x(x-3)}=\frac{x^2-4x-x^2+4x-3}{2x(x-3)}=\frac{-3}{2x(x-3)}$$

There is nothing else that can be factorised on the numerator, so this is the final answer

$$\frac{-3}{2x(x-3)} \text{ or } \frac{3}{2x(3-x)}$$

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